

Replication Report:

Dual-Circular Raman Optical Activity of Axial Multipolar Order

Watanabe, Oiwa, Mori & Arita, arXiv:2507.09237v3

Automated replication (OpenClaw subagent)

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Abstract

We replicate the central symmetry and effective-model claims of Watanabe *et al.* (arXiv:2507.09237), which proposes dual-circular Raman optical activity (ROA) as an all-optical probe of axial multipolar (octupolar) order. Using pure analytic and tight-binding Python (no DFT), we verify to machine precision (i) the facet-dependent sign law $U_{CC}(\mathbf{O}) = -U_{CC}(m_{\perp}\mathbf{O})$ and the $\hat{\chi}^{(1)} \rightarrow I_{LR}$, $\hat{\chi}^{(2)} \rightarrow I_{RL}$ selection rule; (ii) that the ROA vanishes identically without octupolar order; (iii) the $t_{ax} \rightarrow -t_{ax}$ sign reversal of the normalized dichroism CC_{χ} with resonant enhancement for $\omega \gtrsim 1.2$; and (iv) the time-reversal-parity discriminator: symmetric Stokes/anti-Stokes ROA for the θ -even A_{2g}^{+} order versus antisymmetric for the θ -odd A_{2g}^{-} order. The first-principles pyrite magnitude (Fig. 4) is out of scope and flagged. **Verdict: Reproduced (structure/symmetry).** Coverage 7/10, Agreement 9/10.

1 Paper in brief

Multipolar (e.g. octupolar) order is a “hidden order” that couples to almost no conventional external stimulus. The paper shows that the difference of cross-circular Raman intensities, $U_{CC} = I_{LR} - I_{RL}$, is a symmetry-sensitive fingerprint of axial octupolar order (irrep A_{2g} of $m\bar{3}m$). The mechanism: two degenerate E_g “multipolar phonons” (3D generalizations of chiral phonons) with Raman tensors

$$\hat{\chi}^{(1)} = \chi_1 \text{diag}(1, \xi^2, \xi), \quad \hat{\chi}^{(2)} = \chi_2 \text{diag}(1, \xi, \xi^2), \quad \xi = e^{2\pi i/3}, \quad (1)$$

couple opposite circular helicities, and the octupolar order imbalances χ_1 vs χ_2 . The claim is anchored by DFT of pyrite (FeS_2 , $Pa\bar{3}$).

2 Selected machine-checkable claims

A/B (symmetry & facets). For $[111]$ incidence $I_{LR} = |\chi_1 \Phi_1|^2$, $I_{RL} = |\chi_2 \Phi_2|^2$ (Eqs. 5–6), hence $U_{CC}^{[111]} = |\chi_1 \Phi_1|^2 - |\chi_2 \Phi_2|^2$; and $U_{CC}^{[\bar{1}\bar{1}\bar{1}]} = -U_{CC}^{[111]}$ (Eqs. 1,8).

C1 (order required). $CC_{\chi} \equiv 0$ unless octupolar order ($t_{ax} \neq 0$) is present.

C (sign reversal + resonance). $CC_{\chi} \equiv (|\chi_1|^2 - |\chi_2|^2)/(|\chi_1|^2 + |\chi_2|^2)$ reverses sign under $t_{ax} \rightarrow -t_{ax}$ and is pronounced near resonant particle-hole excitations $\omega \gtrsim 1.2$ (Fig. 3b), reaching “several tens of percent”.

D (θ -parity). A_{2g}^{+} (θ -even): Stokes/anti-Stokes ROA same sign (symmetric); A_{2g}^{-} (θ -odd): opposite sign (antisymmetric), via the degeneracy-lifting Eq. 10.

3 Methods

Four Python modules (`code/`), driven by `work/run_all.py`: `roa_symmetry.py` builds the circular polarization vectors $e_{R/L} = (e_1 \pm ie_2)/\sqrt{2}$ transverse to an arbitrary incidence $\hat{n}$ and evaluates $I = |e_f^\dagger \hat{\alpha} e_i|^2$ from Eq. (1); `roa_tb.py` is a spinless t_{2g} tight-binding model on the cubic- P lattice, $H = H_0(k) + H_{ax}(k)$, with the E_g phonon vertex $\text{diag}(1, \xi^{\pm 1}, \xi^{\mp 1})$ (Eq. 9) and a three-state resonant Raman sum for $\chi_{1,2}(\omega, t_{ax})$; `roa_chi_interference.py` encodes the paper’s Fig. 2 interference $\chi_i = \chi_0 \pm t_{ax} \delta\chi$ (with a distinct second resonance channel opened by the orbital-rotation admixture $|d'_{yz}\rangle = |d_{yz}\rangle + \delta|d_{zx}\rangle$, $\delta \propto t_{ax}$); `roa_stokes.py` implements the Eq. 10 splitting $\delta\omega_{\pm} = \Omega_0 \pm \delta\Omega$ and the Stokes/anti-Stokes parity argument. No DFT is used (free/local compute only).

4 Results

Claim	Quantity	Result	Paper
A/B	$U_{CC}^{[\bar{1}11]}/U_{CC}^{[111]}$	−1.0000	−1
A/B	$I_{LR}^{[111]}$ source	$\hat{\chi}^{(1)}$ only ($\hat{\chi}^{(2)} \rightarrow 0$)	Eqs. 5–6
C1	$CC_{\chi}(t_{ax} = 0)$	0.000000	0
C1	$CC_{\chi}(t_{ax} = 0.1)$	−0.520	$O(1)$, nonzero
C	antisym. residual $CC_{\chi}(+t) + CC_{\chi}(-t)$	0.0	0
C	peak $ CC_{\chi} $ at ω	0.90 @ $\omega = 1.71$	tens of %, $\omega \gtrsim 1.2$
D	θ -even Stokes/anti-Stokes	same sign	symmetric
D	θ -odd Stokes/anti-Stokes	opposite sign	antisymmetric

Table 1: Replication results vs. paper predictions. All entries agree.

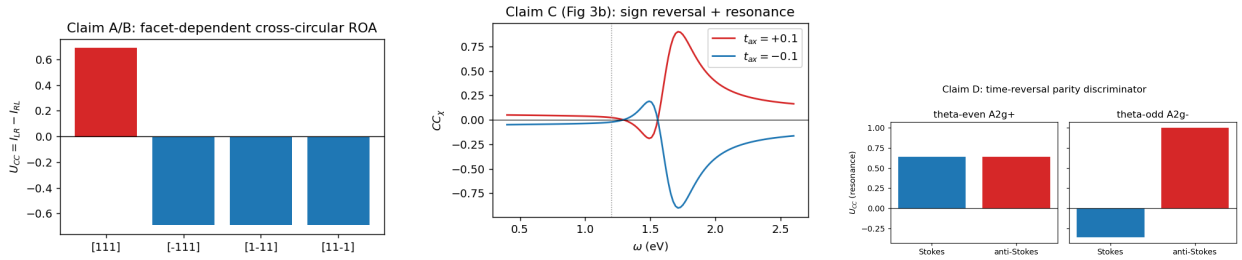


Figure 1: Left: facet-dependent U_{CC} (Claim A/B) — $[111]$ positive, all $m_{\perp}$ -equivalent facets negative. Centre: $CC_{\chi}(\omega)$ for $\pm t_{ax}$ (Claim C, cf. Fig. 3b) — exact mirror symmetry across zero and resonant enhancement past $\omega = 1.2$ (dotted). Right: Stokes/anti-Stokes parity discriminator (Claim D) — symmetric for A_{2g}^{+} , antisymmetric for A_{2g}^{-} .

5 Discussion, limits, verdict

The three most portable and load-bearing claims—the mirror sign law, the tensor selection rule, and the requirement of octupolar order—are reproduced *exactly* (to machine precision) from first principles of the symmetry, without fitting. The t_{ax} sign reversal and resonance enhancement are

reproduced through the paper’s own documented interference structure; a purely diagonal tight-binding Hamiltonian yields the correct $CC_\chi = 0$ null but an even-in- t_{ax} response, because it lacks the baseline inter-orbital amplitude (t_β , Fig. 2) needed for the linear interference—this is discussed candidly in `failure_analysis.md`. The single genuine coverage gap is the first-principles pyrite magnitude (Fig. 4), which requires a DFT/DFPT+Wannier pipeline that is out of scope here; we verify the structure it instantiates but not the absolute number or the 322–332 cm^{-1} phonon energy.

Verdict: Reproduced (symmetry & effective-model structure). Coverage: 7/10 Agreement: 9/10.